**Q1. JEE Main 2026 (21 January Shift 2)**

Let $y^2 = 12x$ be the parabola with its vertex at O. Let P be a point on the parabola and A be a point on the $x$-axis such that $\angle \mathrm{OPA} = 90^\circ$. Then the locus of the centroid of such triangles OPA is :

(1) $y^2 - 2x + 8 = 0$
(2) $y^2 - 9x + 6 = 0$
(3) $y^2 - 4x + 8 = 0$
(4) $y^2 - 6x + 4 = 0$

**Q2. JEE Main 2026 (21 January Shift 2)**

Let one end of a focal chord of the parabola $y^2 = 16x$ be $(16, 16)$. If $\mathrm{P}(\alpha, \beta)$ divides this focal chord internally in the ratio $5 : 2$, then the minimum value of $\alpha + \beta$ is equal to :

(1) 16
(2) 5
(3) 7
(4) 22

**Q3. JEE Main 2026 (28 January Shift 2)**

Let A be the focus of the parabola $y^2 = 8x$. Let the line $y = \mathrm{m}x + \mathrm{c}$ intersect the parabola at two distinct points $B$ and $C$. If the centroid of the triangle $ABC$ is $\left(\frac{7}{3}, \frac{4}{3}\right)$, then $(BC)^2$ is equal to :

(1) 80
(2) 41
(3) 89
(4) 32

**Q4. JEE Main 2026 (24 January Shift 2)**

Let the image of parabola $x^2 = 4y$, in the line $x - y = 1$ be $(y + a)^2 = b(x - c)$, $a, b, c \in$ N. Then $a + b + c$ is equal to

(1) 8
(2) 6
(3) 12
(4) 4

**Q5. JEE Main 2026 (22 January Shift 2)**

Let the locus of the mid-point of the chord through the origin O of the parabola $y^2 = 4x$ be the curve S. Let P be any point on S. Then the locus of the point, which internally divides OP in the ratio $3 : 1$, is :

(1) $2x^2 = 3y$
(2) $3y^2 = 2x$
(3) $2y^2 = 3x$
(4) $3x^2 = 2y$

**Q6. JEE Main 2026 (22 January Shift 1)**

If the chord joining the points $\mathrm{P}_1(x_1, y_1)$ and $\mathrm{P}_2(x_2, y_2)$ on the parabola $y^2 = 12x$ subtends a right angle at the vertex of the parabola, then $x_1x_2 - y_1y_2$ is equal to

(1) 284
(2) 280
(3) 288
(4) 292

**Q7. JEE Main 2026 (23 January Shift 2)**

An equilateral triangle OAB is inscribed in the parabola $y^2 = 4x$ with the vertex O at the vertex of the parabola. Then the minimum distance of the circle having $AB$ as a diameter from the origin is

(1) $2(8 - 3\sqrt{3})$
(2) $2(3 + \sqrt{3})$
(3) $4(6 + \sqrt{3})$
(4) $4(3 - \sqrt{3})$

**Q8. JEE Main 2026 (21 January Shift 1)**

Let O be the vertex of the parabola $x^2 = 4y$ and Q be any point on it. Let the locus of the point P, which divides the line segment OQ internally in the ratio $2:3$ be the conic C. Then the equation of the chord of $C$, which is bisected at the point $(1, 2)$, is :

(1) $5x - 4y + 3 = 0$
(2) $5x - y - 3 = 0$
(3) $4x - 5y + 6 = 0$
(4) $x - 2y + 3 = 0$

## ANSWERS AND SOLUTIONS

**1.** (1) **2.** (3) **3.** (1) **4.** (2) **5.** (3) **6.** (3) **7.** (4) **8.** (1)

**1.** **(1)** Parabola $y^2 = 12x$, $a = 3$. Let $P = (3t^2, 6t)$, $O = (0,0)$, $A = (h, 0)$.

$\angle OPA = 90^\circ \Rightarrow \vec{PO} \cdot \vec{PA} = 0$:

$(-3t^2)(h - 3t^2) + (-6t)(-6t) = 0 \Rightarrow h = 3t^2 + 12$.

Centroid $G = \left(\frac{0+3t^2+h}{3}, \frac{0+6t+0}{3}\right) = (2t^2 + 4,\ 2t)$.

Let $G = (X, Y)$: $t = Y/2$, $X = Y^2/2 + 4$.

Locus: $y^2 - 2x + 8 = 0$.

**2.** **(3)** Parabola $y^2 = 16x$, $a = 4$, focus at $(4, 0)$. Point $(16, 16)$ has parameter $t_1 = 2$.

Other end of focal chord: $t_2 = -1/t_1 = -1/2$, giving $(4 \cdot 1/4, 8 \cdot (-1/2)) = (1, -4)$.

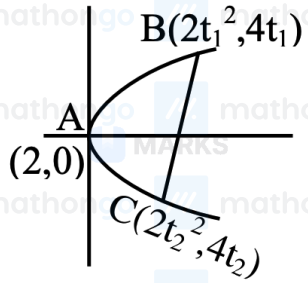
P divides the chord in ratio 5:2 internally (two cases):

From $(16, 16)$ to $(1, -4)$: $P = \left(\frac{37}{7}, \frac{12}{7}\right)$, $\alpha + \beta = 7$.

From $(1, -4)$ to $(16, 16)$: $P = \left(\frac{82}{7}, \frac{72}{7}\right)$, $\alpha + \beta = 22$.

Minimum value of $\alpha + \beta = 7$.

**3.**




**(1)**

Coordinates of centroid of triangle ABC are

$$\frac{2}{3}(t_1^2 + t_2^2 + 1) = \frac{7}{3} \Rightarrow t_1^2 + t_2^2 = \frac{5}{2}$$

$$\frac{4}{3}(t_1 + t_2) = \frac{4}{3} \Rightarrow t_1 + t_2 = 1$$

$$(t_1 + t_2)^2 = t_1^2 + t_2^2 + 2t_1t_2 \Rightarrow t_1t_2 = \frac{-3}{4}$$

$$(t_1 - t_2)^2 = (t_1 + t_2)^2 - 4t_1t_2 = 4$$

$$(BC)^2 = 4(t_1^2 - t_2^2)^2 + 16(t_1 - t_2)^2$$

$$\Rightarrow (BC)^2 = 80$$

**4.** **(2)** Reflection of point $(h, k)$ in line $x - y - 1 = 0$:

$\frac{x-h}{1} = \frac{y-k}{-1} = \frac{-2(h-k-1)}{2} = k - h + 1$.

Image: $(k + 1,\ h - 1)$.

Let $(X, Y) = (k + 1,\ h - 1)$, so $h = Y + 1,\ k = X - 1$.

From $h^2 = 4k$: $(Y + 1)^2 = 4(X - 1)$.

Comparing with $(y+a)^2 = b(x-c)$: $a = 1,\ b = 4,\ c = 1$.

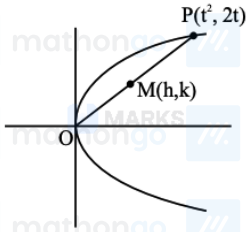
$a + b + c = 1 + 4 + 1 = 6$.

**5.** **(3)** $y^2 = 4x$

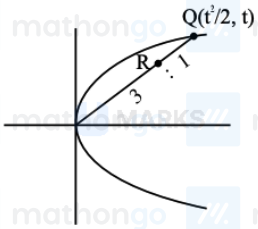
Locus of mid point of OP

$M(h,k) \Rightarrow h = \dfrac{t^2}{2}, k = t$

$\Rightarrow k^2 = 2h \Rightarrow y^2 = 2x$




$S : y^2 = 2x$




$R(h,k)$

$\Rightarrow h = \dfrac{3t^2}{4}, k = \dfrac{3t}{4}$

$t^2 = \dfrac{8h}{3}, t = \dfrac{4k}{3}$

$\Rightarrow \dfrac{16k^2}{9} = \dfrac{8h}{3} \Rightarrow 2k^2 = 3h$

Locus of R : $2y^2 = 3x$

**6.** **(3)** Parabola $y^2 = 12x$ has $4a = 12$, so $a = 3$.

Parametric points: $P_1 = (3t_1^2, 6t_1)$, $P_2 = (3t_2^2, 6t_2)$.

Chord subtends right angle at vertex $O(0,0)$: $\vec{OP_1} \cdot \vec{OP_2} = 0$.

$9t_1^2t_2^2 + 36t_1t_2 = 0$

$\Rightarrow t_1t_2(t_1t_2 + 4) = 0$

$\Rightarrow t_1t_2 = -4$.

$x_1x_2 - y_1y_2 = 9(t_1t_2)^2 - 36(t_1t_2)$

$= 9(16) - 36(-4) = 144 + 144 = 288$.

**7.** **(4)** Parabola $y^2 = 4x$ has vertex $O = (0,0)$. By symmetry, let $A = (t^2, 2t)$, $B = (t^2, -2t)$.

$OA = t\sqrt{t^2+4}$, $AB = 4t$.

For equilateral: $OA = AB \Rightarrow t\sqrt{t^2+4} = 4t \Rightarrow t^2 = 12 \Rightarrow t = 2\sqrt{3}$.

$A = (12, 4\sqrt{3})$, $B = (12, -4\sqrt{3})$.

Circle with $AB$ as diameter: centre $(12, 0)$, radius $= 4\sqrt{3}$.
Minimum distance from origin $= 12 - 4\sqrt{3} = 4(3 - \sqrt{3})$.

**8.** **(1)** Parabola $x^2 = 4y$, vertex O = (0, 0), Q = $(2t, t^2)$ on parabola.
P divides OQ in ratio 2:3: $P = \left(\frac{4t}{5}, \frac{2t^2}{5}\right)$.
Eliminating $t$: $x^2 = \frac{8y}{5}$ (conic C with $a = \frac{2}{5}$).
For chord with midpoint $(h, k)$: $hx - 2ay = h^2 - 2ak$.
At $(1, 2)$: $x - \frac{4y}{5} = 1 - \frac{8}{5} = -\frac{3}{5}$.
$5x - 4y + 3 = 0$.